

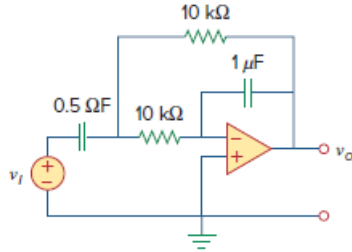
Problem

Obtain the transfer function of the op amp circuit in Fig. 16.112 in the form of

$$\frac{V_o(s)}{V_i(s)} = \frac{as}{s^2 + bs + c}$$

where a , b , and c are constants. Determine the constants.

Figure 16.112:



Step-by-step solution

Step 1 of 2

Using the result of problem 16.14,

$$\frac{V_o}{V_i} = \frac{-Y_1 Y_2}{Y_2 Y_3 + Y_4 (Y_1 + Y_2 + Y_3)}$$

When $Y_1 = SC_1$ $C_1 = 0.5 \mu F$

$Y_2 = \frac{1}{R_1}$ $R_1 = 10 K \Omega$

$Y_3 = Y_2$ $Y_4 = SC_2$ $C_2 = 1 \mu F$

Step 2 of 2

$$\begin{aligned} \frac{V_o}{V_i} &= \frac{\frac{-SC_1}{R_1}}{\frac{1}{R_1^2} + SC_2 \left(SC_1 + \frac{2}{R_1} \right)} = \frac{-SC_1 R_1}{1 + SC_2 R_1 (2 + SC_1 R_1)} \\ &= \frac{-SC_1 R_1}{S^2 C_1 C_2 R_1^2 + S \cdot 2 C_2 R_1 + 1} \\ &= \frac{-S(0.5 \times 10^{-6})(10 \times 10^3)}{S^2 (0.5 \times 10^{-6})(1 \times 10^{-6})(10 \times 10^3)^2 + 2S(1 \times 10^{-6})(10 \times 10^3) + 1} \\ &= \frac{-100S}{S^2 + 400S + 2 \times 10^4} \end{aligned}$$

Therefore

$a = -100$ $b = 400$ $c = 2 \times 10^4$